Activity based

Project Report on

Computer Network Laboratory Project

Submitted to Vishwakarma University, Pune

Under the Initiative of

Contemporary Curriculum, Pedagogy, and Practice (C2P2)

By

Snehal Late

Department of Computer Engineering

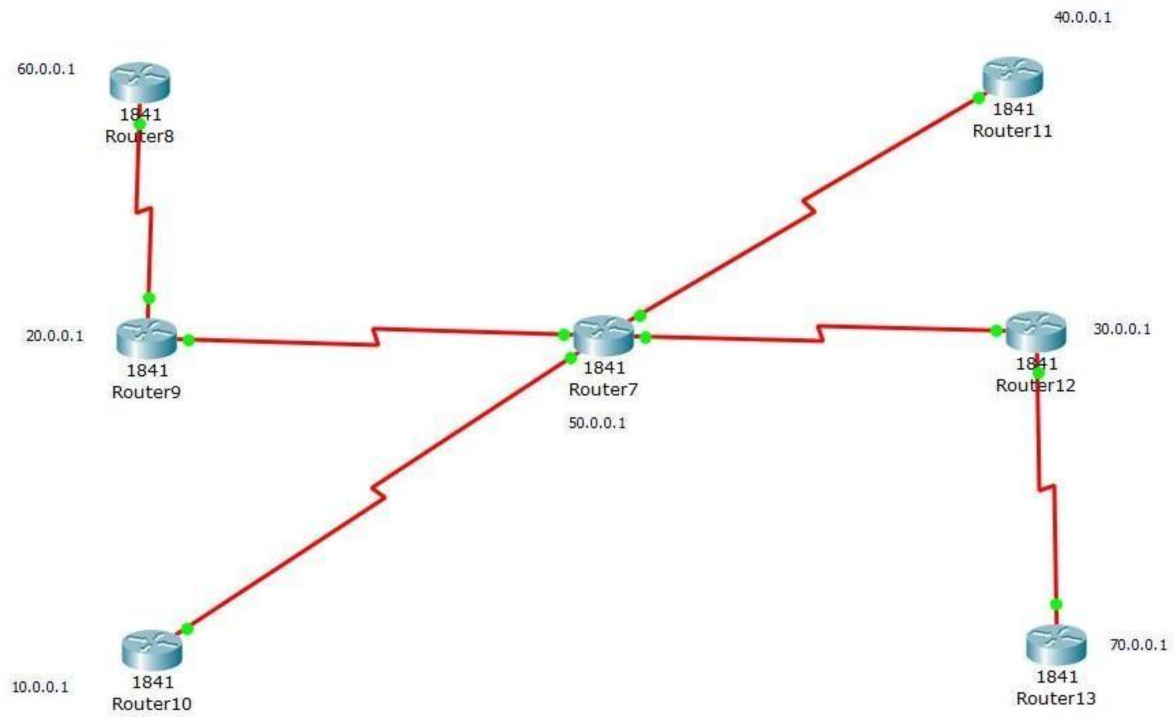
Faculty of Science and Technology

Design and develop a LAN & WAN network and implement static routing

Project Statement:

Implement Static Routing for following network.

Select Class A IP addresses.



Problem Description:

I. Develop a LAN and WAN network for above mentioned diagram.
II. Implement Static routing.
III. Show successful communication in between different Network.

Project Module I:

I. Give detail information about Physical infrastructure which include different types of hardware interfaces used in it. Give detail information about which cable is used.

Hardware Interfaces

1. Router Interfaces:

1. **Serial Interfaces**: Routers often have serial interfaces, especially in enterprise and lab environments, to connect directly to other routers. Serial connections are commonly used for WAN (Wide Area Network) connections between routers over long distances.
2. **Ethernet Interfaces**: Routers also come with Ethernet ports (Fast Ethernet or Gigabit Ethernet) to connect to LAN (Local Area Network) devices such as switches or computers. These interfaces allow data to flow at high speeds within the local network.

2. Switch Interfaces:

1. Switches are designed with multiple Ethernet ports (typically Fast Ethernet or Gigabit Ethernet) that allow multiple devices, like computers and printers, to connect to the network. Switches use these ports to forward data packets within the local network based on MAC addresses.

3. Network Interface Cards (NICs) in Computers:

1. Each computer or device on the network has a Network Interface Card (NIC), which provides a physical port for connecting to the network. The NIC typically has an Ethernet port to connect to the switch using an Ethernet cable. This allows computers to communicate with other devices within the network.

4. WAN Interface Card (WIC):

1. Some routers are equipped with or can support WAN Interface Cards (WICs), which add additional serial or other types of interfaces for WAN connectivity. These are often used to expand the number of connections a router can make, especially for connections to external networks.

Types of Cables Used

1. **Serial Cables (DCE/DTE)**:

    1. Serial cables are commonly used for connections between routers, particularly for WAN links. One end of the cable is connected to the Data Circuit-terminating Equipment (DCE) side, which provides a clock rate, and the other end is connected to the Data Terminal Equipment (DTE) side. This type of connection helps synchronize data flow between routers.

2. **Ethernet Cables**:

    1. **Straight-Through Ethernet Cables**: Used to connect devices of different types, such as connecting a router to a switch or a switch to a computer. Straight-through cables are the most common type for connecting end devices within the LAN.

2. **Crossover Ethernet Cables**: Used to connect two similar devices directly, such as connecting two routers or two switches without using a router or hub in between. These cables are less common in modern setups but still used in specific cases.

3. **Fiber Optic Cables** (Optional):

    1. For longer distances or higher-speed requirements, fiber optic cables can be used. These cables transmit data as light signals and offer much faster speeds and greater distance coverage than copper cables, making them suitable for backbone connections in larger networks.

II. Give detail information of Which IP addresses is used. How many bits allocated for network address and host addresses. Connect computers to the routers. What is the IP address of default gateway in computer. What is the subnet mask.

IP Address Allocation & Subnetting:

IPAddressing Scheme

- **Router8**: 60.0.0.1
- **Router9**: 20.0.0.1
- **Router10**: 10.0.0.1
- **Router11**: 40.0.0.1
- **Router12**: 30.0.0.1
- **Router13**: 70.0.0.1
- **Router7**: 50.0.0.1 (central router connecting all routers)

Each router's network is a Class A IP address, with the network portion defined by the first octet.

Subnetting

- A subnet mask of 255.0.0.0 is used, allocating the first 8 bits for the network address and the remaining 24 bits for host addresses within each subnet.
- This configuration allows each subnet to support a large number of hosts, ensuring scalability.

Network and Host Bits:

- **Network Bits**: 30 bits are used for the network portion of the IP address. This means that the subnet can accommodate only 2 usable IP addresses, perfect for point-to-point connections.
- **Host Bits**: 2 bits are used for the host portion, which means there are only 2 usable IP addresses per subnet. This fits the requirement for connecting just two devices (routers in this case).

Default Gateway and Subnet Mask for Computers:

Subnet Mask for all computers: 255.0.0.0

Default Gateway is the router's LAN IP for each segment:

LAN Segment	Default Gateway
Router8 LAN	60.0.0.1
Router9 LAN	20.0.0.1
Router10 LAN	10.0.0.1
Router11 LAN	40.0.0.1
Router12 LAN	30.0.0.1
Router13 LAN	70.0.0.1

III. Give detail information about following hardware with its role in network

Hardware Components and Their Roles in the Network:

Routers:

- **Model:** Cisco 1841
- **Role:**
    - Routers are the key devices responsible for forwarding data packets between different networks. In this setup, they are configured to use Routing Information Protocol (RIP), which helps them share routing information dynamically. This means that each router can learn about other networks in the system and find the best path for sending data, even if it has to pass through several routers.
    - Each router has serial interfaces that connect to other routers for inter-router communication and Fast Ethernet interfaces that connect to local devices, like computers, within the same network.
- **Configuration:**
    - **Serial Interfaces:** These are used for communication between routers. Each connection between two routers uses a 30-bit subnet mask (255.255.255.252), which is perfect for point-to-point links, providing just two usable IP addresses for the connection.
    - **Fast Ethernet Interfaces:** These interfaces connect the router to local devices (computers or switches) in the network. They typically use a 255.255.255.0 subnet mask to allow devices within the same subnet to communicate.

Switches:

- **Role:**
    - While not specifically detailed in this network setup, switches play a role in connecting multiple devices within the same network. If more local devices were added, switches would help manage the communication between them.

- Switches direct data traffic using MAC addresses, ensuring that the data is sent only to the devices that need it, which makes the network more efficient.

- **Functionality:**

    - Switches help improve network performance by ensuring that traffic is sent only to the right devices, reducing unnecessary load on other devices and ensuring faster communication across the network.

Cables:

1. **Serial DCE/DTE Cable:**

    - **Purpose:** These cables are used for establishing connections between routers over long distances, known as WAN (Wide Area Network) connections.
    - **Clocking:** The DCE (Data Circuit-terminating Equipment) end of the cable provides the clocking signal that's necessary for the serial communication between routers.
    - **Usage:** This type of cable is essential for setting up point-to-point links between routers, which allows them to communicate across different network segments.

2. **Straight-through Ethernet Cable:**

    - **Purpose:** These cables are used to connect routers to local devices (like computers or switches) within the same network.
    - **Usage:** They ensure that devices can communicate within the local area network (LAN) and access the broader network resources.

WAN Interface Card (WIC):

- **Role:** The **WAN Interface Card (WIC)** is an add-on module for routers that provides extra **serial interfaces.**
- **Usage:** For example, with the Cisco 1841 router, using a WIC-2T module allows the router to have multiple serial connections. This is crucial for creating flexible and scalable network designs by connecting more routers over WAN links.

Fast Ethernet:

- **Role:** Fast Ethernet interfaces (such as FastEthernet0/0 and FastEthernet0/1) are used for connecting the router to local devices within the LAN.
- **Functionality:** These interfaces allow routers to assign IP addresses to local devices (like computers or switches) and act as a default gateway, helping devices within the same network communicate with each other.

    - They also ensure that local network traffic is routed efficiently without needing to go through the router's WAN interfaces.

Conclusion :

This project provided practical experience in implementing static routing and configuring a network with routers and switches. I learned how to assign IP addresses, subnet masks, and default gateways for proper communication between devices in a LAN and WAN setup. The hands-on work with routing protocols like RIP, along with configuring serial and Ethernet cables, deepened my understanding of network design and data flow. Overall, the assignment enhanced my knowledge of network configuration and management.

Project Module II

I. Configure all routers with IP addresses

Link	Router Interface	IP Address	Subnet Mask
Router8-Router9	Router8 Serial0/1/0	60.0.0.1	255.0.0.0
	Router9 Serial0/1/0	20.0.0.1	255.0.0.0
Router9-Router7	Router9 Serial0/1/1	20.0.0.1	255.0.0.0
	Router7 Serial0/1/0	50.0.0.1	255.0.0.0
Router10-Router7	Router10Serial0/1/0	10.0.0.1	255.0.0.0
	Router7 Serial0/1/1	50.0.0.1	255.0.0.0
Router11-Router7	Router11Serial0/0/0	40.0.0.1	255.0.0.0
	Router7 Serial0/1/0	50.0.0.1	255.0.0.0
Router12-Router7	Router12Serial0/1/0	30.0.0.1	255.0.0.0
	Router7 Serial0/0/0	50.0.0.1	255.0.0.0
Router13-Router12	Router13Serial0/1/0	70.0.0.1	255.0.0.0
	Router12Serial0/1/1	30.0.0.1	255.0.0.0

II. Configure different routes with Static routing

Static routes are configured to enable manual routing between routers for the network. Static routing requires the network administrator to define specific routes for each router to reach the desired destination. The following static routing configuration uses Class A IP addresses for the network.

Static Routing Configuration

1. Router 7 (Central Router) Configuration:

Static Routing:

Router7(config)# ip route 50.0.0.0 255.0.0.0 10.0.0.2

Router7(config)# ip route 60.0.0.0 255.0.0.0 20.0.0.2

Router7(config)# ip route 70.0.0.0 255.0.0.0 30.0.0.2

2. **Router8 Configuration:**

Static Routing:

Router8(config)# ip route 20.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 30.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 40.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 50.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 60.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 70.0.0.0 255.0.0.0 10.0.0.1

3. **Router 9 Configuration:**

Static Routing:

Router9(config)# ip route 10.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 30.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 40.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 50.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 60.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 70.0.0.0 255.0.0.0 20.0.0.1

4. **Router 10Configuration:**

Static Routing:

Router10(config)# ip route 10.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 20.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 40.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 50.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 60.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 70.0.0.0 255.0.0.0 30.0.0.1

5. **Router 11 Configuration:**

Static Routing:

Router11(config)# ip route 10.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 20.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 30.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 50.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 60.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 70.0.0.0 255.0.0.0 40.0.0.1

6. **Router 12 Configuration:**

Static Routing:

Router12(config)# ip route 10.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 20.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 30.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 40.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 60.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 70.0.0.0 255.0.0.0 50.0.0.1

7. **Router 13 Configuration**

Static Routing

Router13(config)# ip route 10.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 20.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 30.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 40.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 50.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 70.0.0.0 255.0.0.0 60.0.0.1

8. Router 14 Configuration

Static Routing

Router14(config)# ip route 10.0.0.0 255.0.0.0 70.0.0.1

Router14(config)# ip route 20.0.0.0 255.0.0.0 70.0.0.1

Router14(config)# ip route 30.0.0.0 255.0.0.0 70.0.0.1

Router14(config)# ip route 40.0.0.0 255.0.0.0 70.0.0.1

III. Activate different route

After configuring the static routes on each router, the routes will be manually defined and active. To ensure the static routes are functioning properly and that routers can reach all destinations, you can verify the route status and connectivity using specific commands.

show ip route

ping 10.0.0.2

IV. Save all configuration in router

To ensure that the static routing configurations are retained after a router restart, it is essential to save the current running configuration to non-volatile random access memory (NVRAM). This will prevent the router from losing its configuration when it reboots.

write memory

show running-config

V. In project report copy all the codes

1. Router 7 (Central Router) Configuration:

enable

configure terminal

Router7(config)# interface serial 0/0

Router7(config-if)# ip address 10.0.0.1 255.0.0.0

Router7(config-if)# no shutdown

Router7(config)# interface serial 0/1

Router7(config-if)# ip address 20.0.0.1 255.0.0.0

Router7(config-if)# no shutdown

Router7(config)# interface serial 0/2

Router7(config-if)# ip address 30.0.0.1 255.0.0.0

Router7(config-if)# no shutdown

Router7(config)# interface serial 0/3

Router7(config-if)# ip address 40.0.0.1 255.0.0.0

Router7(config-if)# no shutdown

Static Routing:

Router7(config)# ip route 50.0.0.0 255.0.0.0 10.0.0.2

Router7(config)# ip route 60.0.0.0 255.0.0.0 20.0.0.2

Router7(config)# ip route 70.0.0.0 255.0.0.0 30.0.0.2

2. Router8 Configuration:

enable

configure terminal

Router8(config)# interface serial 0/0

Router8(config-if)# ip address 10.0.0.2 255.0.0.0

Router8(config-if)# no shutdown

Static Routing:

Router8(config)# ip route 20.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 30.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 40.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 50.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 60.0.0.0 255.0.0.0 10.0.0.1

Router8(config)# ip route 70.0.0.0 255.0.0.0 10.0.0.1

3. Router 9 Configuration:

enable

configure terminal

Router9(config)# interface serial 0/0

Router9(config-if)# ip address 20.0.0.2 255.0.0.0

Router9(config-if)# no shutdown

Static Routing:

Router9(config)# ip route 10.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 30.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 40.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 50.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 60.0.0.0 255.0.0.0 20.0.0.1

Router9(config)# ip route 70.0.0.0 255.0.0.0 20.0.0.1

4. Router 10Configuration:

enable

configure terminal

Router10(config)# interface serial 0/0

Router10(config-if)# ip address 30.0.0.2 255.0.0.0

Router10(config-if)# no shutdown

Static Routing:

Router10(config)# ip route 10.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 20.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 40.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 50.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 60.0.0.0 255.0.0.0 30.0.0.1

Router10(config)# ip route 70.0.0.0 255.0.0.0 30.0.0.1

5. Router 11 Configuration:

enable

configure terminal

Router11(config)# interface serial 0/0

Router11(config-if)# ip address 40.0.0.2 255.0.0.0

Router11(config-if)# no shutdown

Static Routing:

Router11(config)# ip route 10.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 20.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 30.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 50.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 60.0.0.0 255.0.0.0 40.0.0.1

Router11(config)# ip route 70.0.0.0 255.0.0.0 40.0.0.1

6. Router 12 Configuration:

enable

configure terminal

Router12(config)# interface serial 0/0

Router12(config-if)# ip address 50.0.0.2 255.0.0.0

Router12(config-if)# no shutdown

Static Routing:

Router12(config)# ip route 10.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 20.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 30.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 40.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 60.0.0.0 255.0.0.0 50.0.0.1

Router12(config)# ip route 70.0.0.0 255.0.0.0 50.0.0.1

7. Router 13 Configuration

Router13(config)# interface serial 0/0

Router13(config-if)# ip address 60.0.0.2 255.0.0.0

Router13(config-if)# no shutdown

Static Routing

Router13(config)# ip route 10.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 20.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 30.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 40.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 50.0.0.0 255.0.0.0 60.0.0.1

Router13(config)# ip route 70.0.0.0 255.0.0.0 60.0.0.1

8. Router 14 Configuration

Router14(config)# interface serial 0/0

Router14(config-if)# ip address 70.0.0.2 255.0.0.0

Router14(config-if)# no shutdown

Static Routing

Router14(config)# ip route 10.0.0.0 255.0.0.0 70.0.0.1

Router14(config)# ip route 20.0.0.0 255.0.0.0 70.0.0.1

Router14(config)# ip route 30.0.0.0 255.0.0.0 70.0.0.1

Router14(config)# ip route 40.0.0.0 255.0.0.0 70.0.0.1

Write memory

Conclusion :

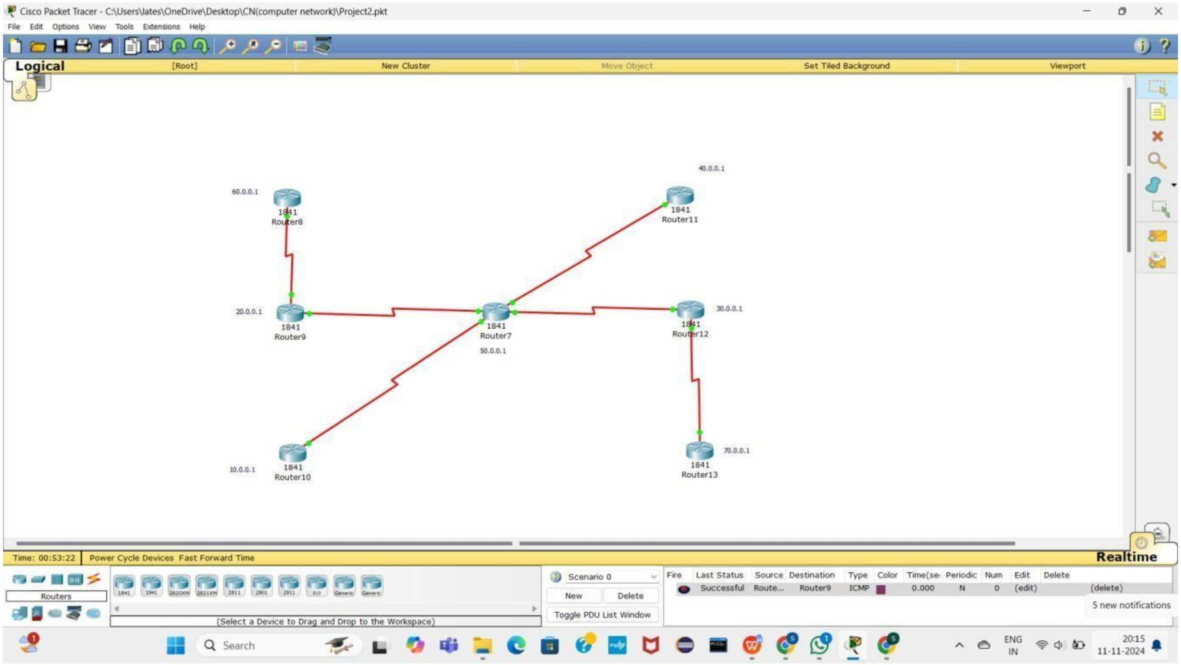
This project taught me how to configure static routing between routers using Class A IP addresses and /30 subnets. I learned to manually define routes with the ip route command and verify connectivity through commands like show ip route and ping. I also understood the importance of saving configurations to NVRAM. Overall, the project enhanced my skills in network routing, configuration, and troubleshooting.

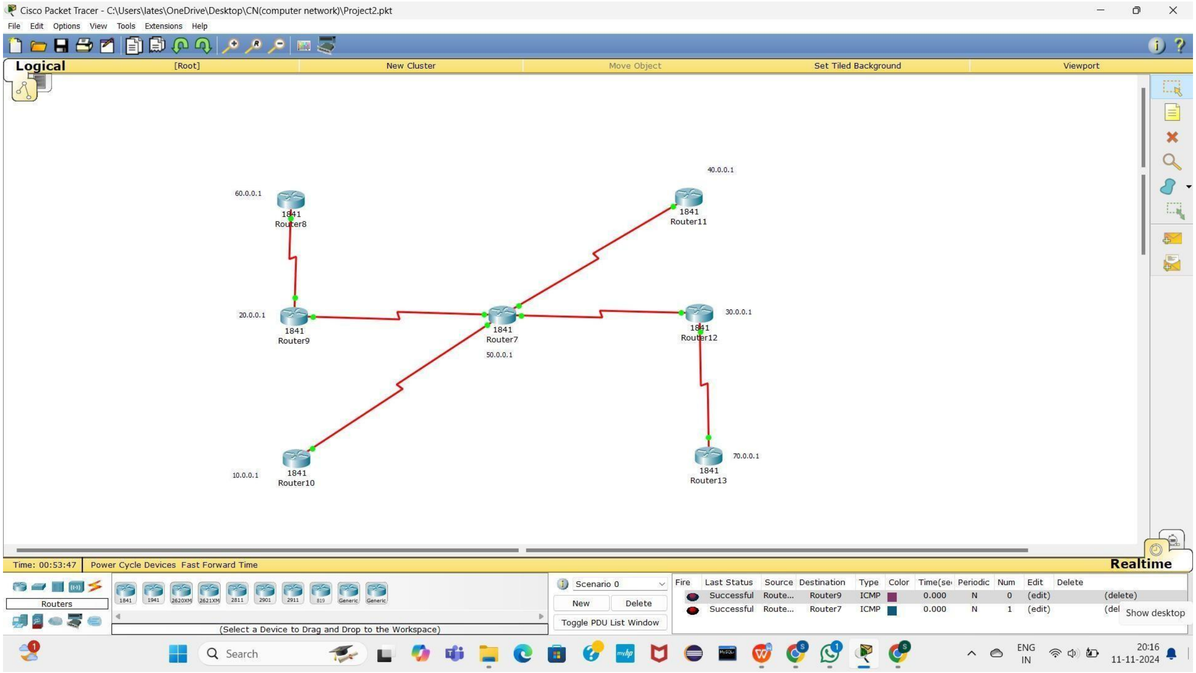
Project Module III

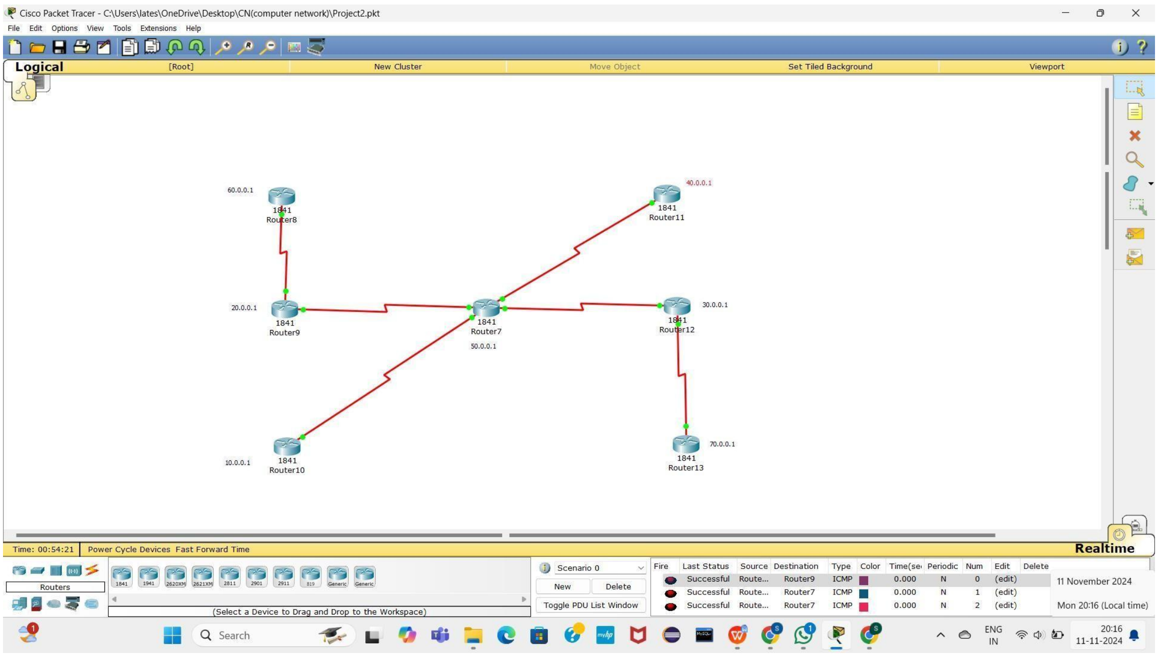
I. Show screenshot of communication in between different Network.

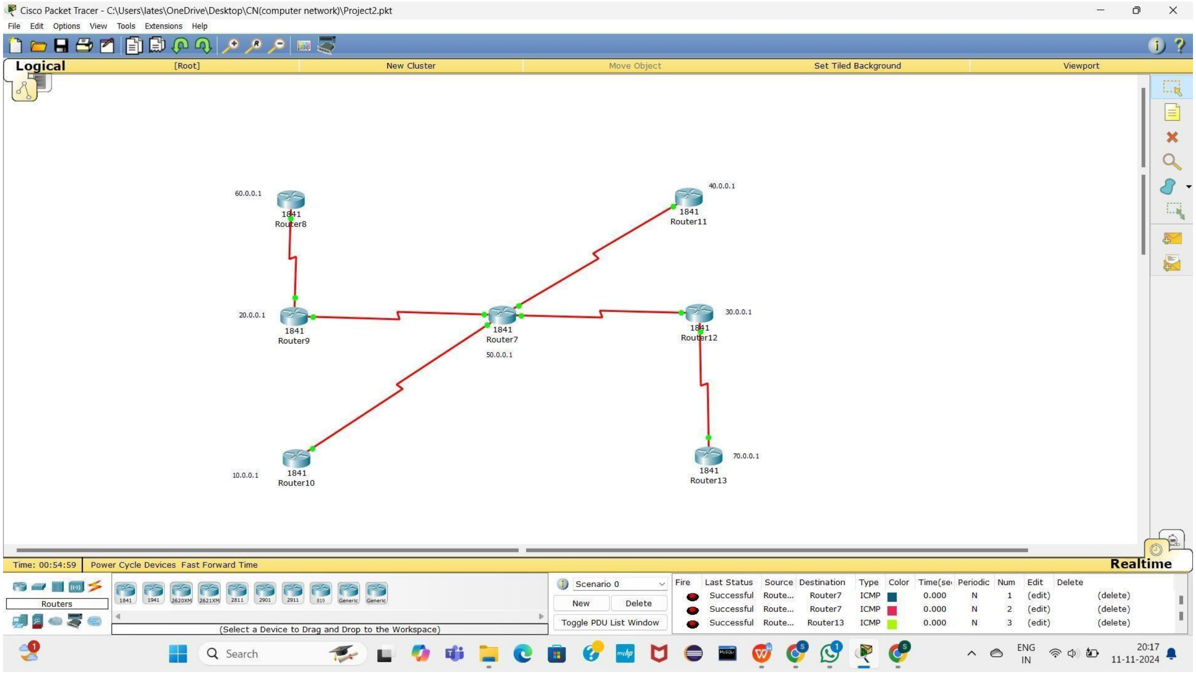
Separate screenshot of communication in between two different network.

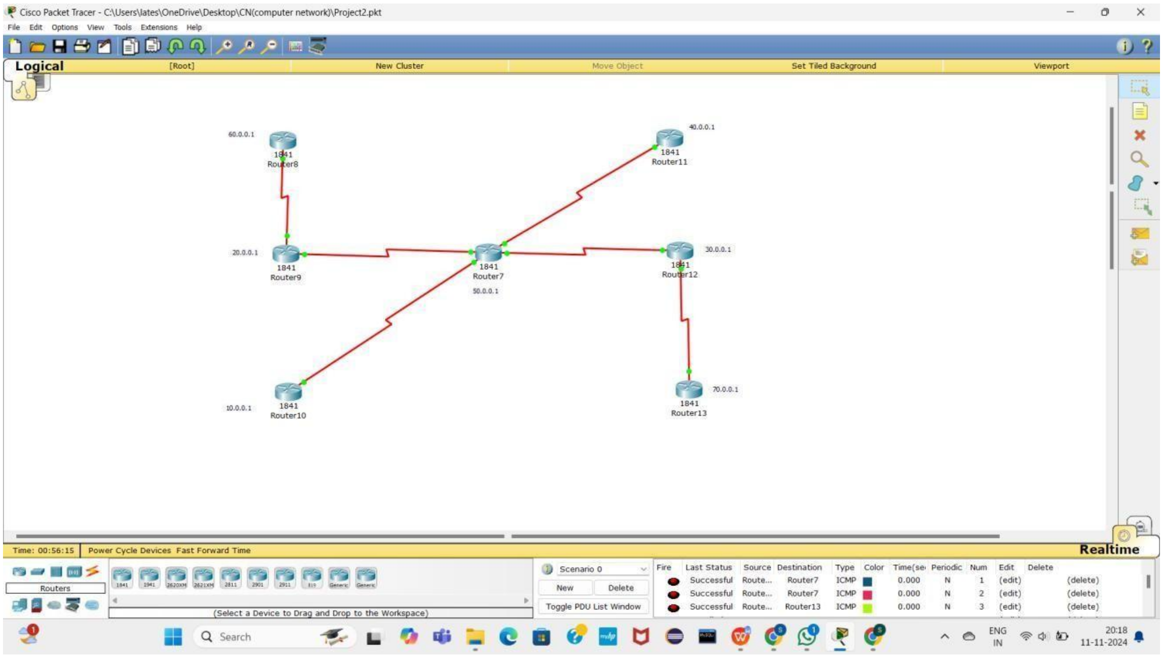
Create a report with all screenshot




60.0.0.1
Router8
40.0.0.1
Router11
20.0.0.1
Router9
Router7
50.0.0.1
30.0.0.1
Router12
10.0.0.1
Router10
70.0.0.1
Router13


60.0.0.1
Router8
40.0.0.1
Router11
20.0.0.1
Router9
Router7
50.0.0.1
30.0.0.1
Router12
10.0.0.1
Router10
70.0.0.1
Router13

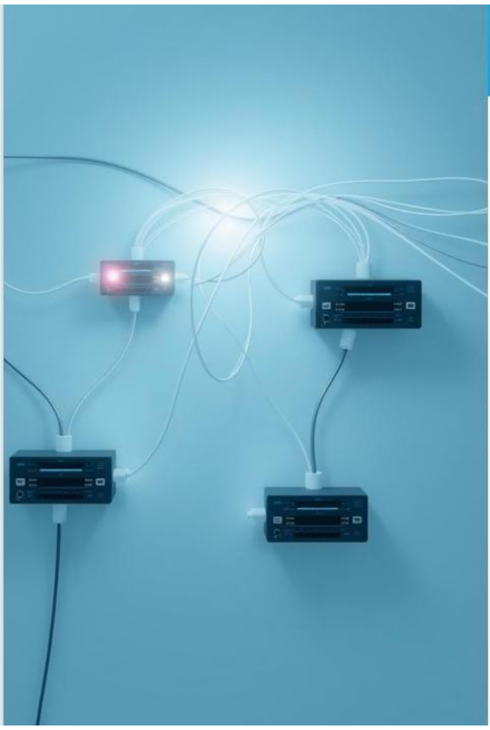




Conclusion :

This project helped me understand the process of implementing static routing in a network using Class A IP addresses. I learned how to configure static routes between different routers and successfully establish communication between networks. The hands-on experience allowed me to gain practical knowledge in troubleshooting, verifying connectivity, and saving router configurations. Overall, the assignment enhanced my skills in network configuration and routing.

PPT



PROJECT 2

Static Routing

Static routing is a network configuration method where network administrators manually define the paths data packets should take to reach their destination. This contrasts with dynamic routing, where routers automatically learn and adapt routes based on network traffic.

PROJECT 2

NAME : Snehal Late

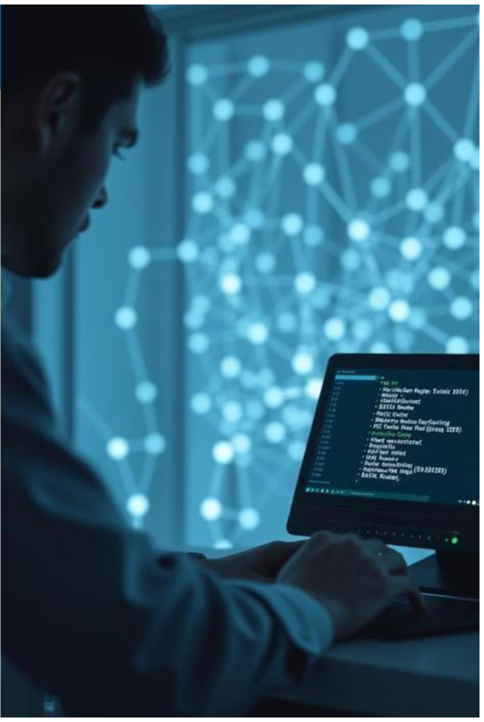
SRN : 202201577

DIV : D

Roll No : 30

Subject : CN

What is Static Routing?



Static routing involves manually configuring each route on a router. Network administrators define the specific paths that data packets should follow to reach destinations. This approach provides precise control over network traffic flow.

1 Manual Configuration

Network administrators define specific routes manually, providing precise control over data flow.

2 Fixed Paths

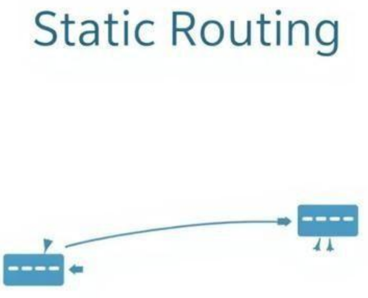
Routes remain constant unless manually changed, ensuring predictable traffic flow.

3 Simple Networks

Best suited for small, stable networks with predictable traffic patterns.

Characteristics of Static Routing

Static routing offers several distinct characteristics that differentiate it from dynamic routing. These characteristics are crucial for understanding its strengths and limitations.



Simplicity

Easy to configure and manage, particularly in small, straightforward networks.

Predictability

Routes are fixed, ensuring consistent and predictable data flow.

Security

Offers a higher degree of security, as routes cannot be altered by external factors.

Advantages of Static Routing

Static routing offers several advantages, particularly in specific network scenarios. These advantages stem from its deterministic nature and manual configuration.

Reduced Network Overhead

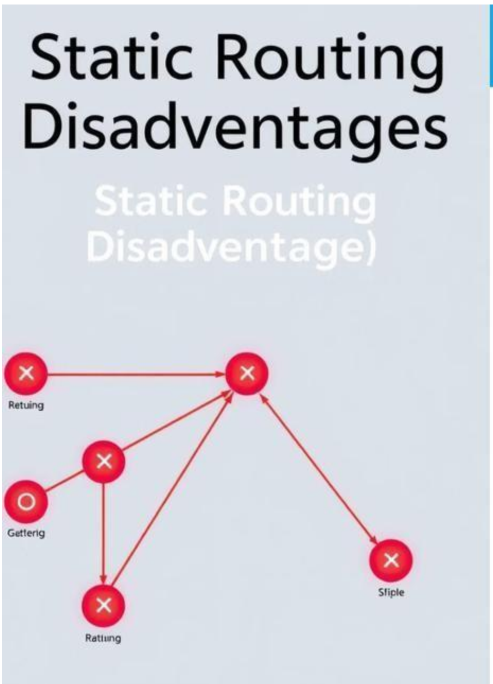
Less processing power required for route calculations, minimizing network latency.

Increased Security

Routes are manually configured, preventing external factors from influencing data flow.

Cost-Effective for Small Networks

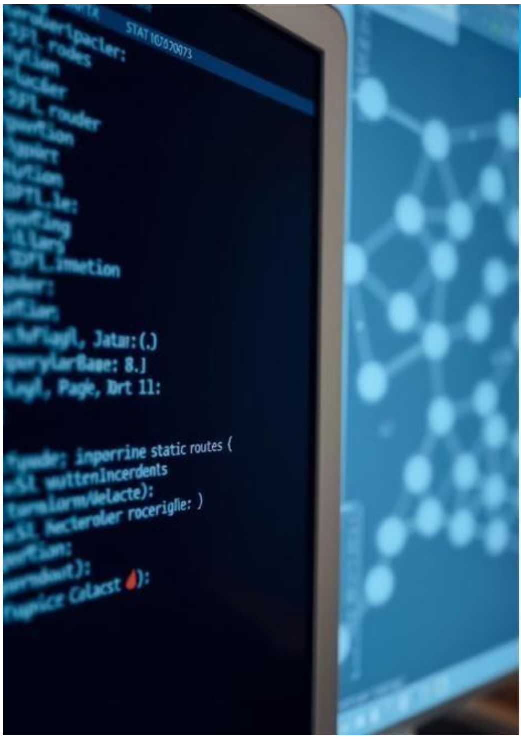
Minimal configuration and management overhead reduces the cost of network operation.



Disadvantages of Static Routing

Static routing has certain limitations that make it unsuitable for larger or dynamic networks. These disadvantages arise from its manual configuration and fixed route paths.

Scalability	Difficult to manage in large networks with numerous routes.
Flexibility	Limited ability to adapt to network changes, leading to potential routing issues.
Management Complexity	Requires manual updates for any network changes, increasing administrative workload.



Configuring Static Routes

Configuring static routes involves manually defining the paths data packets should take to reach their destination. This process typically involves specifying the network address, next hop, and interface.

1. **Identify the Network**

   Determine the network address and subnet mask of the destination network.

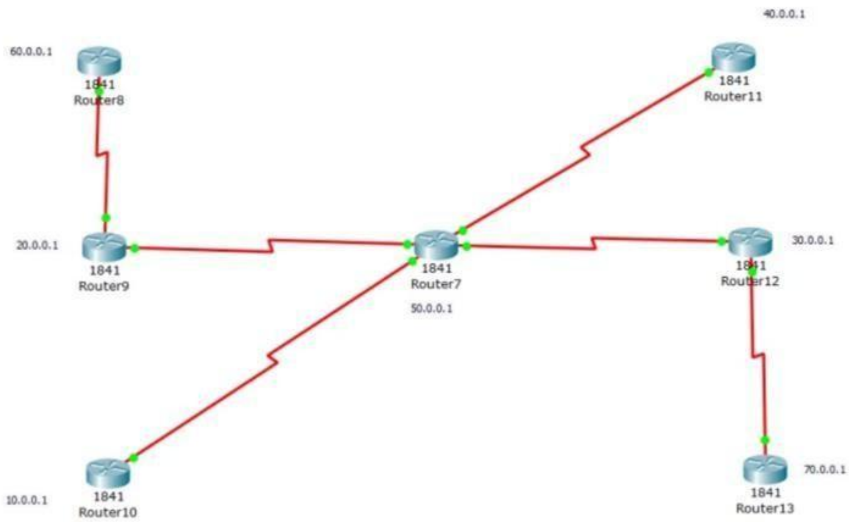
2. **Specify the Next Hop**

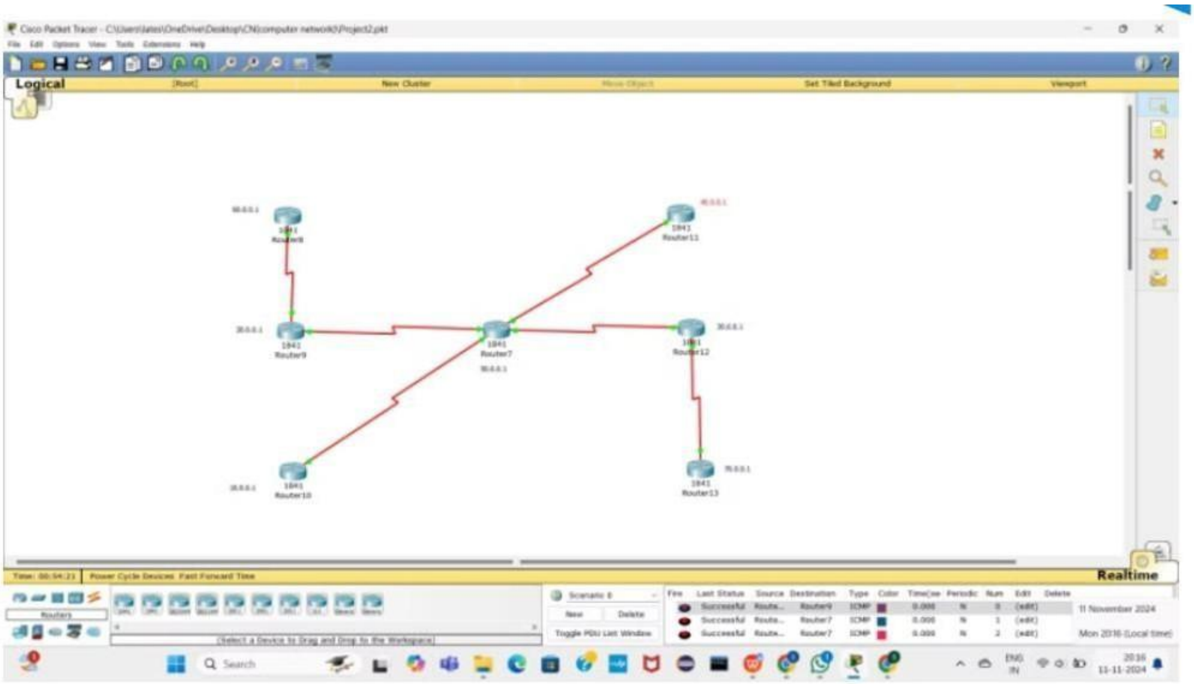
   Define the next router or device that data packets should be forwarded to.

3. **Choose the Interface**

   Select the interface on the router that will be used to send the data packets.

Static Routing

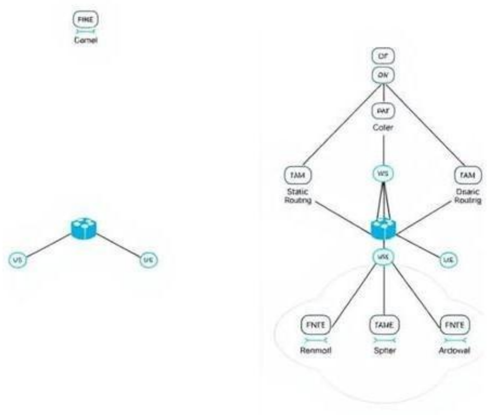




Configure all routers with IP addresses

Link	Router Interface	IP Address	Subnet Mask
Router8-Router9	Router8 Serial0/1/0	60.0.0.1	255.0.0.0
	Router9 Serial0/1/0	20.0.0.1	255.0.0.0
Router9-Router7	Router9 Serial0/1/1	20.0.0.1	255.0.0.0
	Router7 Serial0/1/0	50.0.0.1	255.0.0.0
Router10-Router7	Router10 Serial0/1/0	10.0.0.1	255.0.0.0
	Router7 Serial0/1/1	50.0.0.1	255.0.0.0
Router11-Router7	Router11 Serial0/0/0	40.0.0.1	255.0.0.0
	Router7 Serial0/1/0	50.0.0.1	255.0.0.0
Router12-Router7	Router12 Serial0/1/0	30.0.0.1	255.0.0.0
	Router7 Serial0/0/0	50.0.0.1	255.0.0.0
Router13-Router12	Router13 Serial0/1/0	70.0.0.1	255.0.0.0
	Router12 Serial0/1/1	30.0.0.1	255.0.0.0

Routing Camparison



Conclusion and Key Takeaways

Static routing is a simple and secure approach for managing network traffic flow in smaller, stable networks. It offers predictable routing, reduced overhead, and cost-effectiveness. However, for larger and more dynamic networks, dynamic routing is generally preferred due to its flexibility and adaptability.